Md. Irtiza Hossain

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LinkedIn GitHub

Research Interests

Computer Vision, Deep Learning, Scene Understanding, Behaviour Recognition in Surveillance, Graph Neural Networks for Visual Reasoning, Uncertainty-Aware 3D Object Detection, Knowledge Distillation, and AI Applications in Security and Autonomous Systems.

Education

Master of Science in Computer Science and Engineering *Jun 2025 – Present*
BRAC University, Dhaka, Bangladesh

Bachelor of Science in Computer Science and Engineering *Dec 2019 – May 2024*
BRAC University, Dhaka, Bangladesh

- **CGPA:** 3.90 / 4.00
- **Thesis:** *Enhanced CNN Approaches for Multi-Image Embedding in Image Steganography*
- **Honours:** Vice Chancellor's List (5 semesters); Dean's List (2 semesters)

Publications

Published

- **Edge-Native Digitization of Handwritten Marksheets: A Hybrid Heuristic-Deep Learning Framework** 2026
  Md. Irtiza Hossain, Junaid Ahmed Sifat, Abir Chowdhury, Md. Imran Bin Azad
  IEEE QPain 2026 (Accepted)
  arXiv — Code
- **Enhanced CNN Approaches for Multi-Image Embedding in Image Steganography** 2024
  Md. Irtiza Hossain, Samiul Kadir, Farhan Ishraq Fagun, Ishtiaq Samiul, Rafi Zaman Saukhin
  ICICT 2024
  10.1109/ICICT64387.2024.10839645 — Code

Under Review & In Preparation

- **GLE-SSD-Net: Uncertainty-Aware Distillation for LiDAR-Camera 3D Detection**
  Sami Sadat, **Md. Irtiza Hossain**, Shaownak Md. Ibne Shahriar Talukder, Md. Golam Rabiul Alam
  In Preparation
- **SmokeGraph-Bench: A Benchmark for Smoking Behaviour Detection with Geometric Relational Baselines**
  Md. Irtiza Hossain, Sami Sadat, Junaid Ahmed Sifat, Md. Waseq Alauddin Alvi, Md. Khalilur Rahman
  In Preparation
- **Uncertainty-Aware and Efficient 3D Object Detection for Autonomous Driving: A Comprehensive Survey**
  Sami Sadat, **Md. Irtiza Hossain**, Md. Golam Rabiul Alam
  In Preparation

Teaching Experience

Adjunct Lecturer *Nov 2024 – Present*
BRAC University, Dhaka, Bangladesh

- Delivered lectures and supervised laboratory sessions across seven courses: Data Structures & Algorithms, Artificial Intelligence, Machine Learning, Cryptography, Computer Interfacing & Microprocessors, Database Management Systems, System Analysis & Design, and Introduction to Robotics.

Undergraduate Teaching Assistant *Oct 2022 – May 2024*
BRAC University, Dhaka, Bangladesh

- Assisted instruction in *Programming Language II (Object-Oriented Programming)* (Feb 2023 – May 2024) and *Differential Calculus and Coordinate Geometry* (Oct – Dec 2022); held regular office hours and graded examinations.

Industry Experience

Junior Software Engineer *Aug 2024 – Nov 2024*
Orbit Informatics, Dhaka, Bangladesh

- Architected and delivered a production Chrome extension with a React frontend and Node.js backend, enabling dynamic browser integration for enterprise users.

Trainee Engineer (.NET) *Jun 2024 – Aug 2024*
PRAN-RFL Group (othoba.com), Dhaka, Bangladesh

- Automated the abandoned-cart detection pipeline, contributing to a measured 1–2 % increase in recovered sales revenue through targeted follow-up campaigns.

Selected Projects

- **CLEAR-MoE++** Code
  Developed a post-training sparse Mixture-of-Experts pipeline for Vision Transformers using selective expert extraction, optimized routing, and runtime-aware dispatch for efficient inference.
- **Explainable AI for Brain Cancer Classification from MRI Scans** Code
  Applied Grad-CAM and SHAP attribution methods on CNN classifiers to localise tumour regions and improve model transparency for clinical review.
- **Telecom Churn Prediction** Code
  Built an ensemble ML pipeline (XGBoost, Random Forest, SHAP explanations) to identify at-risk subscribers from usage and billing features.
- **Power X Gym — Full-Stack Web Application (MERN Stack)** Live
  Designed and deployed a gym management platform with role-based authentication, class scheduling, and payment integration.

Technical Skills

Research & ML Frameworks: PyTorch, TensorFlow, Keras, Scikit-learn, OpenCV, Detectron2, MMDetection, Hugging Face Transformers

Programming Languages: Python (primary), C, C#, JavaScript, PHP

Tools & Platforms: Git, Docker, Linux/Bash, LaTeX, Jupyter, Weights & Biases, Google Colab / CUDA-enabled GPU clusters

Web & Backend: React, Node.js, Redux, MongoDB, MySQL, Firebase, .NET, PHP

References

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Dr. Muhammad Iqbal Hossain
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